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# Matt Raymond

*Generative ML + Nanochemistry*

I develop principled methods for composing pretrained generative models to solve grand challenges in computational nanochemistry

## Education

**Ph.D. in Signal and Image Processing and Machine Learning** — Ann Arbor, MI
*ECE Dept., University of Michigan* *(GPA 3.92)* — 08/22-**04/27**

- Open source PyTorch/CUDA code for all papers available on GitLab: gitlab.eecs.umich.edu/mattrmd-public/
- Co-advised by Drs. Angela Violi (MechE) and Clayton Scott (ECE)
- Thesis topic: **Generative Nanoparticle Design**

**M.Sc. in Computer Science** — Ann Arbor, MI
*CSE Dept., University of Michigan* *(GPA 3.91)* — 08/20–04/22

**B.Sc. in Computer Science** — Orange, CA
*Chapman University* *(GPA 3.86, magna cum laude)* — 08/16–05/20

## Research Experience

**Directed Study & Research** — Ann Arbor, MI
*Drs. Angela Violi (MechE) and Clayton Scott (ECE)* — 01/21–present

- Develop algorithm to sample phase-separating proteins from pretrained diffusion models without retraining
- >10× accelerated screening/simulation throughput during collaborations with chemists and plasma physicists
- Perform 20× sparser selection of chemical features with a novel, multi-task approach using boosted trees
- Supervise undergrad/masters research in geometric deep learning, deep Gaussian processes, and active learning
- Co-author grant proposals, from small departmental grants to multi-million dollar research centers

**Directed Study & Research** — Ann Arbor, MI
*Dr. Mania Meibody (Architecture & Urban Planning)* — 07/20–11/21

- Used Babylon.js, Node.js, React, and Redis to develop a metaverse-like pedagogical tool for architecture

**Directed Study & Research** — Orange, CA
*Drs. Eric Linstead and Justin Walsh* — 01/20–05/20

- Used facial recognition to map the interpersonal dynamics of astronauts for the ISS Archeology project

## Work Experience

**Data Science Intern** — Troy, MI
*Toyoda Gosei North America* — 05/20

- Hired, but the internship was canceled because of COVID-19

**Instrument Programmer** — Long Beach, CA
*Lotus Instruments* — 09/19–11/19

- Implemented controls and data transfer protocols for government-contracted gas chromatography instruments

**Software Engineering Intern** — Troy, MI
*Toyoda Gosei North America* — 05/19–08/19

- Saved 2k man-hours and $60k per year through automated purchase order tracking with ASP.NET
- Collaborated with Cost Management to solidify requirements and return on investment

## Technical Skills

PyTorch, CUDA, Slurm, Git, Docker, Python, Jax/Flax, LaTeX, WandB, diffusion models, geometric deep learning, interdisciplinary collaboration, transformers, optimization, linear algebra, probability theory, equivariant neural networks, active learning, probabilistic modeling

## Journal Articles

**M. Raymond**, J. C. Saldinger, P. Elvati, and A. Violi, "Universal feature selection for simultaneous interpretability of multitask datasets," *Journal of Cheminformatics*, vol. 18, no. 1, p. 23, 2026. 10.1186/s13321-025-01096-z, https://gitlab.eecs.umich.edu/mattrmd-public/gendesc/gendesc-main-repo

**M. Raymond**, P. Elvati, J. C. Saldinger, J. Lin, X. Shi, and A. Violi, "Machine learning models for Si nanoparticle growth in nonthermal plasma," *Plasma Sources Science and Technology*, vol. 34, no. 3, p. 35014, 2025. 10.1088/1361-6595/adbae1, https://gitlab.eecs.umich.edu/mattrmd-public/ntp-silicon

J. C. Saldinger, **M. Raymond**, P. Elvati, and A. Violi, "Domain-agnostic predictions of nanoscale interactions in proteins and nanoparticles," *Nature Computational Science*, vol. 3, no. 5, pp. 393–402, 2023. 10.1038/s43588-023-00438-x, https://gitlab.eecs.umich.edu/violigroup/ml/neclas/-/releases

## Conference Proceedings

**M. Raymond**, P. Elvati, and A. Violi, "Mapping Kinetic Bottlenecks in Soot Precursor Growth via Graph Analysis of Stochastic Trajectories," in *41st International Symposium on Combustion*, 2026. Accepted

## Workshops

**M. Raymond**, Y. Zhu, J. Zhang, A. Violi, and C. Scott, "Joint Diffusion Sampling via Positive-Unlabeled Guidance for Multi-Modal Data," in *ICML 2025 Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences*, 2025. URL: https://openreview.net/forum?id=1o89PN84NH

**M. Raymond**, A. Violi, and C. Scott, "Joint Optimization of Piecewise Linear Ensembles," in *IEEE International Workshop on Machine Learning for Signal Processing*, 2024. 10.1109/MLSP58920.2024.10734791, https://gitlab.eecs.umich.edu/mattrmd-public/joplen-repositories/joplen/

## Invited Talks

**Joint Diffusion Sampling via Positive-Unlabeled Guidance for Multimodal Data** — Ann Arbor, MI
*University of Michigan, Michigan Student Symposium for Interdisciplinary Statistical Sciences* — 03/26

**From Proteins to Nanoparticles: Domain-Agnostic ML for the Nanoscale** — Ann Arbor, MI
*University of Michigan, Tools and Technology Seminar Series* — 03/25

**Joined Diffusion Models for Nanoparticle Generation** — Orange, CA
*Chapman University, Fowler School of Engineering Seminar Series* — 11/24

**Machine Learning Models for Nanoparticle Growth in Nonthermal Plasmas** — Minneapolis, MN
*University of Minnesota, Dusty Plasma Workshop* — 05/24

## Poster Presentations

**Joint Diffusion Sampling via Positive-Unlabeled Guidance for Multi-Modal Data** — Vancouver, CA
*ICML, 2025 Workshop on Foundation Models for Life Science* — 07/25

**Joined Diffusion Models for Nanoparticle Design** — Chicago, IL
*University of Chicago, Midwest Machine Learning Symposium* — 06/25

**Joined Diffusion Models for Nanoparticle Design** — Ann Arbor, MI
*University of Michigan, 2nd Conference on Foundation Models and AI Agents for Science* — 05/25

**Accel. Drug Discovery: Modeling Nanoscale Interactions in Protein-Nanoparticle Systems** — Ann Arbor, MI
*University of Michigan, DATA Spring 2025 Industy Advisory Board meeting* — 03/25

**Machine Learning for Knowledge Transfer in Nanomedicine** — Ann Arbor, MI
*University of Michigan, e-Health and Artificial Intelligence* — 09/24

**Joint Optimization of Piecewise Linear Ensembles** — London, UK
*Imperial College London, IEEE Machine Learning for Signal Processing Workshop* — 09/24

**Joint Optimization Significantly Improves Gradient Boosting** — Minneapolis, MN
*University of Minnesota, Midwest Machine Learning Symposium* — 05/24

★ **Best Poster Award** Press Release Official Link

**ML Boosts Efficiency in Atomic Nonthermal Plasma Simulations** — Minneapolis, MN
*University of Minnesota, Dusty Plasma Workshop* — 05/24

**Joint Optimization of Linear Ensembles** — Ann Arbor, MI
*University of Michigan, Michigan Student Symposium for Interdisciplinary Statistical Sciences* — 03/24

⋆ **Best Poster Nominee**

**A Taste of Your Own Medicine: Tracing Butyrate Production In The Gut** — Ann Arbor, MI
*University of Michigan, e-Heath and AI Student Symposium* — 09/23

**Preventative Healthcare via Accessible Data Tools** — Orange, CA
*Chapman University, Student Scholar Symposium* — 05/20

## Awards and Honors

**Top 15 Poster Presentation** — Minneapolis, MN
*University of Minnesota, Midwest Machine Learning Symposium* — 04/24

- Official Link Press Release

**Excellence in ECE Honor Roll** — Ann Arbor, MI
*University of Michigan* — 05/24

- For managing the SPEECS graduate student seminar during the 2023-2024 academic year: Press Release

**Music Minor Award** — Orange, CA
*Chapman University* — 05/20

- In recognition of outstanding contributions to the conservatory (*i.e.*, tutoring)

**Provost List, 8 semesters** — Orange, CA
*Chapman University* — 08/16–05/20

- $\geq$3.800 term GPA while enrolled in ≥12 graded credits: Source

## Grants and Funding

**J. Robert Beyster Computational Innovation Graduate Fellowship** — Ann Arbor, MI
*University of Michigan* — 07/24

- Tuition, stipend, and health insurance for a year
- "Cutting-edge research linking high-performance computing to applications of societal importance." Press

**e-HAIL Dataset Creation Funding ($5,000 grant)** — Ann Arbor, MI
*University of Michigan* — 11/25

**e-HAIL Summer Student Support Funding ($12,000 stipend)** — Ann Arbor, MI
*University of Michigan* — 01/24

**e-HAIL Summer Student Support Funding ($12,000 stipend)** — Ann Arbor, MI
*University of Michigan* — 01/23

**Chancellor's Scholarship, 8 semesters** — Orange, CA
*Chapman University* — 08/16–05/20

## Reviewing

*Information Fusion (IF 15.5)* — 02/26

*NeurIPS 2025 Workshop on Foundation Models for Life Science* — 09/25

*NeurIPS 2025 Workshop on AI for Science* — 09/25

*ICML 2025 Workshop on Foundation Models for Life Science* — 06/25

## Teaching and Mentorship

**Graduate Student Instructor** — Ann Arbor, MI
*University of Michigan* — 08/25–Present

- Hold office hours and supplemental lectures each week
- Collaboratively write exam problems and teaching materials
- Contribute math animations (*a lá* 3Blue1Brown) for a new course under development

- Principles of Mach. Learning ( EECS 453 ), Prob. Methods in Engineering ( EECS 301 )

**Programming Instructor** — Irvine, CA
*Coding Minds Academy* — 07/20–11/20

**STEM Tutor** — Orange, CA
*Chapman University* — 09/19–05/20

- Group tutoring for Calculus 1-3, Discrete Math, Visual Programming, Assembly Language, Data Structures and Algorithms, Programming Languages, Genetic Biology, and Molecular Genetics

**Supplemental Instructor for Data Structures and Algorithms** — Orange, CA
*Chapman University* — 09/19–12/19

- Held office hours and supplemental lectures each week

**Music tutor** — Orange, CA
*Chapman University* — 01/17–05/20

- Private tutoring for Music Theory, Music History, Aural Skills, and Music Technology
- Led group review sessions for music history

## Service

**Proofreader for a Linear Algebra Textbook** — Ann Arbor, MI
*Cambridge University Press* — 05/23–09/23

- Proofread a linear algebra textbook from Cambridge University Press for Drs. Jeffery Fessler and Raj Rao
- Independently verified proofs, suggested improvements for clarity, and caught LaTeX errors

**Coordinator, Signal Processing in EECS (SPEECS) seminar** — Ann Arbor, MI
*University of Michigan, SPEECS* — 08/23–05/24

**Coordinator, Machine learning theory reading group** — Ann Arbor, MI
*University of Michigan, ML Theory Reading Group* — 01/22–02/22

**Planning Committee Member, Engineering Research Symposium (ERS)** — Ann Arbor, MI
*University of Michigan, ERS* — 07/20–02/21

**Coordinator, COVID-19 Study Group** — Virtual
*Chapman University* — 05/21–07/21

- Coordinated an MIT OpenCourseWare study group for students who lost summer internships due to COVID-19
- Operating Systems , Microeconomic Theory and Public Policy , and Probability Theory

**Vice-President, FPV Drone Club** — Orange, CA
*Chapman University* — 01/20–05/20

## Class Projects

**The Implicit Bias of Gradient Descent on Separable Multiclass Data** — Ann Arbor, MI
*U-M Course: EECS 598, 559* — 12/22, 05/23

- Developed a proof sketch for the implicit bias of gradient descent on separable data with multiclass PERM losses
- Showed numerically that our conjecture holds for certain well-known loss functions

**Real-Time Distributed Learning in Connected & Autonomous Vehicles (CAVs)** — Ann Arbor, MI
*U-M Course: EECS 571* — 12/21

- Designed distributed learning protocol for sparse gradient propagation
- Implemented simulated learning environment in Tensorflow
- Demonstrated superior generalization, with fewer assumptions than Federated Learning

**Domain Exploration Through Artificial Curiosity** — Ann Arbor, MI
*U-M Course: EECS 545* — 12/20

- Developed simulated Martian terrain for training and evaluation
- Implemented Shmidhuber's definition of artificial curiosity using convolutional auto-encoders
- Performed evaluation against naive models to illustrate effectiveness of artificial curiosity

**Needlecast: On-the-Fly Reconfiguration of Spacecraft Flight Software** — Ann Arbor, MI
*U-M Course: EECS 587* — 12/20

- Collaborated with NASA staff to draft specifications for protocols

- Designed a library for booting NASA core Flight System (cFS) applications on-the-fly
- Implemented Needlecast (see here ) as a plug-and-play header file for NASA cFE
- Developed a simulated network switch and web interface for straightforward debugging

**Preventative Healthcare via Accessible Data Tools** Orange, CA
*Chapman Course: CPSC 353* 05/20

- Developed a web application to rank establishments by the number of COVID-19 cases in their area
- Utilized REST APIs and front-end web development to develop a user-friendly interface

**AI-Driven Contemporary Archaeology for the International Space Station** Orange, CA
*Chapman Course: CPSC 393* 01/20

- Analyzed project requirements with Dr. Justin Walsh (co-PI of ISS Archeology )
- Used facial recognition to map the interpersonal dynamics of astronauts